

FairHire: AI-Driven Resume Profiling and Technical Interview Automation

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Abstract—In today's competitive job market, technical hiring processes often face challenges related to biases and inefficiencies, leading to inconsistent and sometimes unfair candidate evaluations. FairHire is designed to address these challenges by creating a fairer, more objective, and technologically advanced interview process. The platform begins by analyzing candidate resumes using Natural Language Processing (NLP) algorithms to extract critical skills and qualifications, which inform the creation of tailored, role-specific interview questions. By leveraging large language models (LLMs), FairHire adapts questions dynamically, generating contextually relevant and progressively detailed inquiries based on the candidate's responses. This approach ensures a nuanced and thorough evaluation, closely aligned with both the candidate's background and the job's requirements. During the interview, FairHire captures video and audio responses in real time, applying machine learning and NLP techniques to score each answer on criteria such as relevance, accuracy, and depth, thereby minimizing human biases and ensuring a consistent, data-driven assessment. The platform aggregates individual response scores to provide a comprehensive evaluation of technical skills, enabling recruiters to make well-informed hiring decisions efficiently. By combining technological innovation with a commitment to fairness, FairHire aims to transform technical recruitment, promoting inclusivity, quality, and consistency in candidate evaluation.

Keywords: Resume Profiling, Interview Automation, Natural Language Processing (NLP), Skill Extraction, Technical Question Generation, Candidate Scoring, Fair Hiring

I. INTRODUCTION

In today's job market, companies are increasingly focused on fair and inclusive hiring practices to build diverse and effective teams. However, traditional recruitment processes often rely on human judgment, which can unintentionally

introduce biases. These biases—whether unconscious or intentional—may influence hiring decisions based on factors unrelated to a candidate's qualifications, like first impressions or assumptions. Research shows that around 75% of employers have regretted hiring decisions due to such biases, which not only affect candidates individually but also limit companies' access to diverse talent. This highlights the need for more objective and consistent approaches to hiring.

The demand for skilled technical professionals, such as software engineers and data scientists, has also increased significantly, leading to a surge in applications for each job. Recruiters are now responsible for reviewing high volumes of resumes and conducting many rounds of interviews, a process that is both time-consuming and difficult to standardize. Under time pressure, even qualified candidates may be overlooked, and it can be challenging to fairly assess every candidate. This adds to the complexity and often results in inconsistent evaluations, making hiring decisions less effective and reducing the overall quality of recruitment.

The current recruitment process is therefore limited by biases, heavy workloads, and inconsistent candidate experiences, which all contribute to inefficiencies. Personal biases in decision-making can result in talented candidates—especially those from diverse backgrounds—being overlooked. Additionally, the sheer volume of applications often overwhelms recruiters, who may not have enough time or resources to evaluate each candidate thoroughly. As a result, companies struggle to hire the right talent efficiently, and the candidate experience suffers.

FairHire aims to address these challenges by automating

key parts of candidate profiling and interview evaluation. Using advanced technologies like NLP and ML, FairHire helps reduce subjectivity, standardize the interview process, and provide fair, data-driven assessments. The platform begins by analyzing resumes with NLP to identify key skills, experiences, and qualifications. It then generates interview questions that are tailored to each candidate's background, adapting dynamically as the candidate responds, creating a more in-depth and relevant interview experience.

During interviews, FairHire captures video and audio responses in real time and evaluates them for relevance, accuracy, and depth using ML and NLP techniques. This automated scoring is consistent and objective, ensuring that each candidate is assessed fairly. By combining and analyzing these scores, FairHire provides recruiters with a comprehensive overview of each candidate's abilities. This approach not only enhances fairness in the hiring process but also makes recruitment more efficient, helping companies make informed, data-driven decisions while promoting a more inclusive hiring experience.

II. LITERATURE STUDY

The first paper in this literature survey presents an AI-based interview simulator designed to improve the interviewing skills of novice requirements engineers, addressing a gap in traditional RE courses. Using the Design Science methodology, the simulator provides simulated interview scenarios for practicing different types of questions. Powered by IBM Watson, it processes user inputs and retrieves relevant information from requirements documents, offering immediate feedback to refine questioning techniques. An exploratory study showed that students using the simulator performed better in real interviews. The system also incorporates ambiguity to simulate real-world scenarios. Future work could enhance the simulator with advanced AI techniques like NLP and machine learning.[1]

Interactive Interview Chatbot introduces an AI-powered system designed to automate the candidate selection process. Using Natural Language Processing (NLP), the chatbot conducts interviews, reducing biases and fatigue associated with human interviewers. The system verifies candidate identity, asks role-based questions, and analyzes responses to provide immediate feedback and scores. Benefits include improved efficiency, lower costs, and easy updates with new questions. Challenges such as NLP limitations, scoring biases, speech recognition issues, and privacy concerns regarding video monitoring are acknowledged. The research supports AI chatbot integration in recruitment, while stressing the need to address these limitations for fairness and effectiveness.[2]

Interview Bot with Automatic Question Generation and Answer Evaluation addresses challenges in recruitment, particularly resume screening and candidate evaluation. It proposes a solution using Natural Language Processing (NLP) to extract structured data and programming keywords from resumes. The paper highlights the dual challenge of generating technical questions and evaluating responses using NLP. It

advocates for assessing candidates' technical skills not only through MCQs but also through short answer questions for deeper insights. The system matches candidates to suitable roles based on resume data and includes a proctoring system to ensure interview integrity. The scoring mechanism combines job fit with interview performance for a comprehensive candidate assessment, improving the efficiency and accuracy of hiring practices.[3]

Artificial Intelligence Chatbots are New Recruiters explores the role of AI chatbots in streamlining the recruitment process. The system begins with candidates uploading their resumes, which the chatbot analyzes using natural language processing (NLP) to extract key skills, qualifications, and experiences. This data is then compared to the job role requirements for an initial suitability assessment. The chatbot dynamically generates personalized interview questions based on each candidate's profile, adapting the complexity of questions as the interview progresses. Speech recognition technology is integrated to convert spoken responses into text, which are then evaluated and scored against predefined criteria. This approach ensures consistency in question delivery and scoring, reducing bias and promoting fairness. The system also analyzes patterns in candidates' responses, providing recruiters with data-driven insights for informed decision-making. At the end of the process, a detailed report is generated, summarizing the candidate's performance, strengths, and areas for improvement. [4]

Robotic Process Automation for Recruitment Process explores how RPA automates tasks like resume screening, candidate communication, interview scheduling, and onboarding. Using Natural Language Processing (NLP), RPA bots match candidates' resumes to job requirements and engage with them via platforms like email and WhatsApp. This automation speeds up the hiring process, reduces costs, and minimizes errors. RPA also streamlines onboarding and enhances candidate matching with machine learning. Predictive analytics helps recruiters build proactive talent pipelines. While RPA improves efficiency, it lacks the ability to assess soft skills, though future AI advancements may address this gap.[5]

Interview Bot Development with Natural Language Processing and Machine Learning describes the creation of an AI-driven interview bot designed to automate competency assessments using NLP and Machine Learning within the Behavioral Event Interview (BEI) framework. The bot analyzes candidate responses with NLP techniques, including tokenization and part-of-speech tagging. Machine learning models, trained on past interview data, evaluate responses using Bayesian inference and Term Frequency (TF). The bot generates open-ended questions based on the STAR method and adjusts complexity in real-time. It provides unbiased evaluations, speech-to-text processing, and proctoring to ensure authenticity. After the interview, it generates a detailed competency report. The system reduces bias, improves scalability, and enhances the efficiency of competency-based in-

interviews. Future enhancements may include voice interaction and psychological adaptability to increase its applicability in HR settings. [6]

Efficient Resume-Based Re-Education for Career Recommendation in Rapidly Evolving Job Markets by Ashrafi et al. (2023) presents an AI-powered framework designed to help professionals adapt to rapidly changing job markets. By analyzing resumes using Natural Language Processing (NLP), the system identifies key skills and experiences, mapping them to current market trends and job requirements. It also predicts future skill demands with Machine Learning (ML), offering personalized educational recommendations to address skill gaps. The system provides efficient, scalable career guidance, though it faces challenges like reliance on accurate resume data and concerns over algorithmic bias. Despite these issues, the paper demonstrates AI and ML's potential in enhancing professional development in evolving job markets.[7]

Implicit Skills Extraction Using Document Embedding and Its Use in Job Recommendation by Gugnani and Misra (2020) introduces a novel method for extracting implicit skills from job descriptions and resumes using document embedding techniques. By leveraging word and sentence embeddings, the model identifies skills that are implied rather than explicitly stated, improving traditional skill extraction processes. This approach enhances job recommendation systems by capturing a wider range of relevant skills, thereby improving candidate-job matching accuracy. While the method offers significant improvements, it faces challenges related to data quality, resource-intensive computations, and implementation complexity. Overall, the paper emphasizes the potential of advanced NLP techniques for more effective job recommendation systems.[8]

A System for Detecting Professional Skills from Resumes Written in Natural Language by Chifu et al. (2017) presents a framework designed to efficiently extract professional skills from resumes written in natural language. Using Natural Language Processing (NLP) and pattern recognition algorithms, the system identifies and classifies both explicit and implicit skills across various resume formats. By leveraging skill categories and linguistic pattern analysis, the framework provides a comprehensive view of a candidate's qualifications. The system enhances recruitment efficiency by automating skill identification, reducing manual effort for recruiters. However, challenges such as handling ambiguous language, synonyms, and context-dependent presentations remain. The paper emphasizes the importance of NLP in improving skill detection for more informed hiring decisions.[9]

Multimodal Speech Emotion Recognition Using Audio and Text by Yoon, Byun, and Jung (2018) introduces a novel framework that combines audio and text data to enhance emotion recognition in speech. Traditional methods focus on either audio features (e.g., pitch, tone) or text features (e.g., sentiment analysis), often missing the full range of emotional expression. This study utilizes a multimodal approach, in-

tegrating Mel-frequency cepstral coefficients (MFCCs) for audio analysis and word embeddings for text analysis. By merging these data sources, the system significantly improves emotion detection accuracy, particularly for emotions like joy, sadness, and anger. The results demonstrate better recognition compared to single-modality techniques, making the approach useful in applications such as human-computer interaction and mental health monitoring. However, challenges remain in data integration and contextual understanding, which require further research to overcome.[10]

Speech-To-Text Conversion (STT) System Using Hidden Markov Model (HMM) by Mon and Tun (2015) presents a method for converting spoken language into text using Hidden Markov Models (HMM), a statistical technique well-suited for speech recognition. The system consists of three components: feature extraction, HMM modeling, and decoding. During feature extraction, Mel-frequency cepstral coefficients (MFCCs) are used to capture the speech signal's acoustic features. The HMM models the temporal dynamics of speech, considering variations in pronunciation and intonation. The decoding phase uses algorithms like Viterbi to identify the most likely text sequence from the speech input. The study demonstrates the system's high accuracy, making it suitable for real-time applications. However, challenges include the need for large training datasets and sensitivity to noise and environmental factors. This research highlights the potential of HMM-based systems in improving speech-to-text technology for voice-activated interfaces, transcription, and accessibility applications.[11]

Video-Based Action Detection for Online Exam Proctoring in Resource-Constrained Settings by Felsing, Hal-loluwa, and Fonseka (2023) proposes a solution for online exam proctoring in environments with limited resources. The study introduces an action detection framework that uses video analysis to monitor student behavior during exams, identifying suspicious activities such as cheating or using unauthorized materials. The approach utilizes lightweight machine learning models, optimized for low-spec devices, ensuring accessibility in resource-constrained settings with limited computational power and bandwidth. The framework effectively detects a range of actions while minimizing false positives. The study also addresses privacy and consent concerns, advocating for transparency in proctoring practices. This research highlights the potential of video-based action detection as a reliable and scalable solution for online proctoring, balancing technological innovation with practical considerations for equitable assessment in diverse educational contexts.[12]

We conducted an in-depth study on implementing deep-fake video generation techniques to create realistic and engaging question-asking scenarios during interviews. This approach aims to enhance the candidate's experience by simulating a natural, human-like interaction.[13][14]

III. METHODOLOGY

The methodology of the FairHire system revolves around creating an AI-driven platform that streamlines the recruitment process while minimizing bias. To better understand the flow and interaction of various modules, the following architecture diagram illustrates the system's components and their interactions:

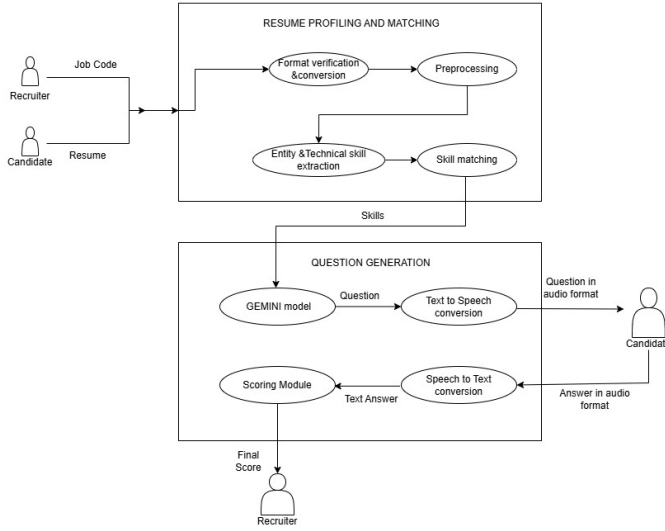


Fig. 1. System Architecture

the system's methodology can be described through the following steps:

- Resume Profiling module
- Question Generation & Scoring module
- Speech Module
- Proctoring Module

A. Resume Profiling

This module extracts key information from candidates' resumes using NLP techniques such as Named Entity Recognition (NER). It identifies essential skills, qualifications, and experience relevant to the job role. This extracted data forms the foundation for the subsequent question generation module, ensuring that the interview questions are tailored to the candidate's qualifications.

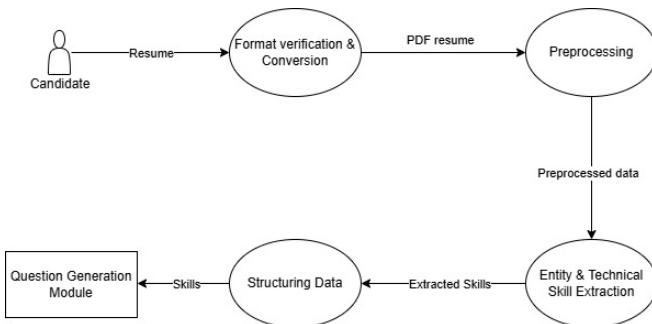


Fig. 2. Resume profiling module

- Resume (Input) is parsed and processed.
- NER identifies key entities.
- Skills, experience, and qualifications are extracted.
- The skills are matched with the key skills needed for a job role and passed to the Question Generation module.

The Resume Profiling Module in FairHire processes resumes using PyPDF2, pdfplumber, and python-docx for text extraction. Text data is tokenized and cleaned to remove noise. spaCy with custom NER models is used for skill extraction, with fuzzywuzzy applied for similarity matching to improve recognition of variant skill names. Extracted skills are further matched with the job description to identify relevant competencies. Regular expressions are employed for extracting name, email, and phone number, with validation checks to ensure data accuracy. Extracted data is structured in JSON format for integration with other modules, while try-except blocks manage parsing errors for improved reliability.

B. Question Generation & Scoring module

The Question Generation and Scoring Modules in FairHire both utilize the Gemini model through prompting (zero-shot or few-shot learning). For question generation, the model is prompted to create dynamic, role-specific interview questions tailored to the candidate's skills, with complexity increasing progressively based on prior responses. For scoring, the Gemini model evaluates candidate answers by analyzing relevance, accuracy, and depth using structured prompts, assigning scores on a predefined scale. This prompt-based approach ensures flexible, adaptive questioning and consistent evaluation without the need for fine-tuning.

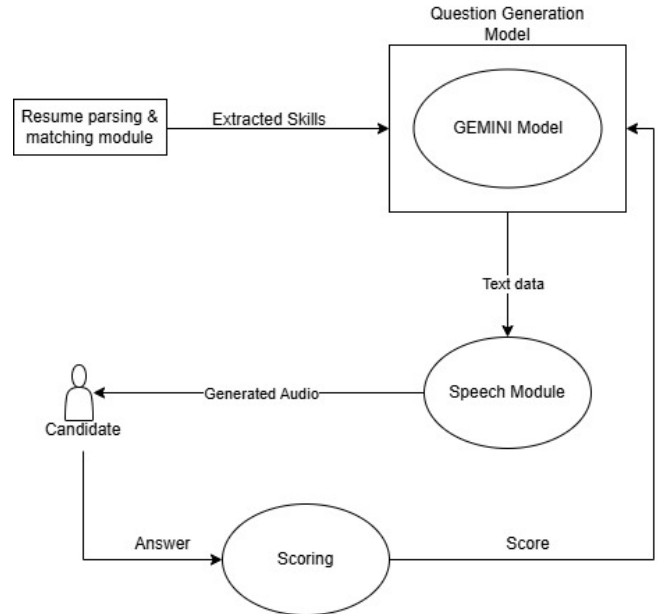


Fig. 3. Question Generation & Scoring module

The Question generation Module in FairHire is a Web-Socket server designed to simulate a human interviewer by dynamically generating tailored questions and scoring

candidate responses. It leverages the Gemini-2.0-flash-exp model from google.generativeai for both question generation and scoring. The module receives inputs such as job_name, candidate_id, num_questions, and skills via WebSocket and prompts the model to select relevant skills for the specified role. If skill parsing fails, it defaults to all provided skills.

The Question Generation logic iterates through selected skills, generating questions at three levels — Basic, Intermediate, and Hard — based on score progression thresholds. Each response is scored on a 0-10 scale for accuracy, relevance, and completeness, with adjustments for vague answers (e.g., “I don’t know”). Timeout handling includes a 120-second limit followed by a 30-second warning period.

Final scores are calculated as a percentage of the actual maximum raw score, with skill-wise scores calculated separately. Results are sent to the client and updated in Django via aiohttp. The system maintains a persistent connection using asyncio with a 120-second timeout and 30-second ping/pong intervals to ensure stable communication.

To encourage candidates to provide detailed answers rather than short, one-word responses, we integrated NLP-based feedback. This system dynamically prompts candidates to elaborate on their answers, ensuring richer responses that allow the evaluation model to assess relevance, accuracy, and depth more effectively.

By adapting question complexity based on candidate performance, managing timeouts, and handling errors gracefully, the Question generation Module efficiently delivers an adaptive, real-time interview experience.

C. Speech Module

The Speech Module is responsible for real-time verbal interactions, converting spoken responses into text using speech-to-text technology. It also converts the generated questions back into speech, ensuring candidates clearly hear them. The module employs advanced NLP techniques to enhance transcription accuracy, accommodates diverse speech patterns, and monitors responses for evaluation and scoring.

- Speech input is captured from the candidate during the interview.
- The Speech-to-Text (STT) system converts spoken responses into text.
- NLP techniques are applied to enhance transcription accuracy and understanding.
- The system also converts generated interview questions from text to speech for the candidate to hear clearly.
- The converted text is passed to the Scoring Module for evaluation based on response content, accuracy, and clarity.

The Text-to-Speech (TTS) system uses the SpeechSynthesis API to vocalize questions from currentQuestion, with text displayed progressively via setInterval. The isSpeaking state controls the avatar animation, which stops on speechRef.current.onend, enabling the microphone for responses.

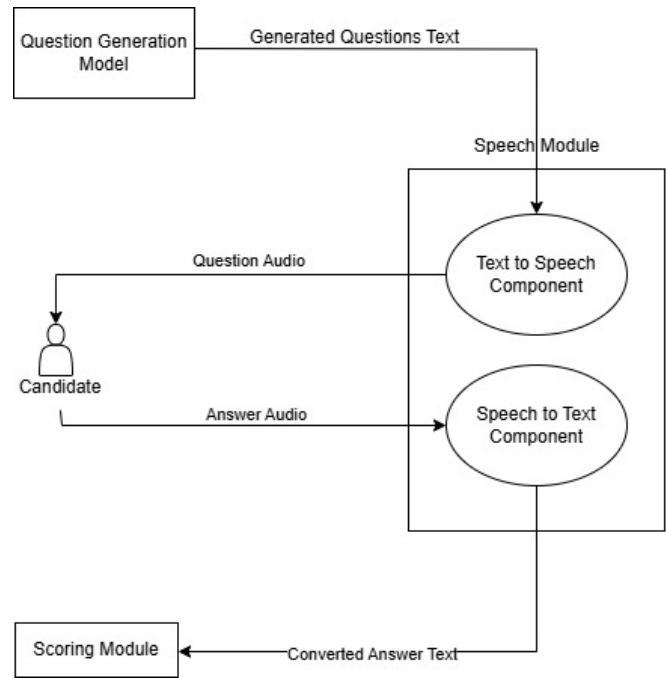


Fig. 4. Speech module

The STT system uses the SpeechRecognition API with continuous and interim results enabled. The instance in recognitionRef.current captures responses via onresult, updating candidateAnswer in real-time. Recording starts after TTS completion and auto-restarts on onend. The sendAnswer function stops recognition upon submission. WebSocket integration and state management ensure synchronization, while visual cues like SpeakingIcon and ListeningIcon enhance interaction.

D. Proctoring Module

The Proctoring Module in FairHire is a Flask microservice that leverages Mediapipe Face Mesh and YOLOv5 for real-time violation detection. The system processes video frames received via POST requests to the proctor endpoint, where frames are decoded using cv2.imdecode. The detect_faces function utilizes face_mesh.process to detect and count faces (up to 5) via multi_face_landmarks, identifying violations when no face or multiple faces are present. The detect_objects function employs yolov5s.pt for object detection, identifying labels such as "cell phone" or "laptop" with a confidence threshold of 0.5.

A proctoring_flag is set based on detected violations, with a corresponding reason dictionary providing detailed boolean flags for each violation type. The system responds with a JSON object containing the candidate_id, proctoring_flag, and reason, ensuring efficient integration with the React frontend, which sends frames every 3 seconds via the captureAndSendFrame function. Robust error handling is implemented using try-except blocks to manage decoding, detection, or processing failures, ensuring reliable operation under varied conditions.

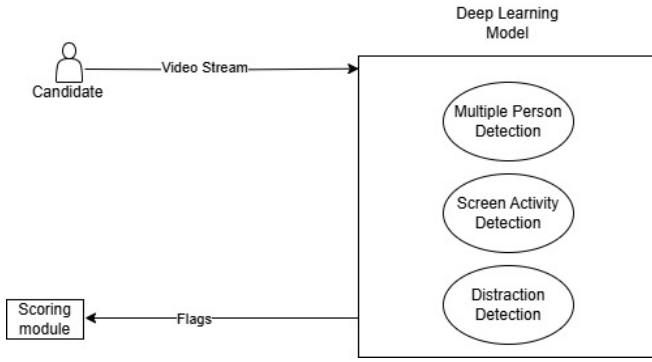


Fig. 5. Proctoring Module

- Detects tab switching during the interview to ensure the candidate remains focused on the interview.
- Identifies and flags multiple faces in the video feed to prevent unauthorized assistance.
- Detects the presence of mobile phones to ensure candidates do not use them for cheating.
- Monitors eye contact to ensure the candidate is engaged with the interview.
- Verifies the candidate's identity by matching their face with the image submitted during the resume profiling.

IV. CHALLENGES FACED

The FairHire system faces challenges in ensuring data accuracy, maintaining real-time performance, and integrating complex modules like speech recognition, proctoring, and scoring. Adapting to resource constraints and accurately interpreting diverse candidate responses also play a crucial role in its effectiveness.

- **Microservice Architecture and Communication** Managing independent microservices using `asyncio`, `websockets`, and `aiohttp` introduced complexities in ensuring loose coupling, stable inter-service communication, and secure data exchange via structured JSON payloads.
- **Real-Time WebSocket Synchronization** Implementing WebSockets for dynamic interactions required handling concurrent connections, managing `asyncio` event loops, and ensuring synchronization between Qgen, speech processing, and scoring. Adding timeout handling and reconnection logic improved stability during network disruptions.
- **Prompt Engineering and Model Behavior** Since FairHire relies on prompting with the `Gemini-2.0-flash-exp` model, crafting prompts that dynamically adjust question difficulty, assess relevance, and handle vague responses required extensive testing and refinement to ensure consistency.
- **Dependency and Environment Management Challenges** Managing dependencies for different libraries, such as `YOLOv5`, `Mediapipe`, and `spaCy`, required isolating each module into its own dedicated virtual environment to prevent conflicts. Each microservice was set up and

run in its respective environment, ensuring compatibility. However, this approach introduced challenges in configuring and maintaining multiple environments, as well as handling their integration with the backend efficiently.

V. EXPERIMENTAL RESULTS

The FairHire project leverages the Gemini model through prompting to develop an AI-driven interview system that enhances objectivity and minimizes bias. Prompting was chosen over fine-tuning to ensure greater flexibility, faster adaptation, and reduced resource requirements. The system dynamically generates tailored interview questions and evaluates candidate responses based on relevance, accuracy, and depth. Evaluation results confirm that FairHire effectively produces relevant questions and delivers consistent, accurate scoring.

The Resume Profiling Module in FairHire was evaluated by comparing its extracted data with manually annotated resumes. The evaluation focused on identifying key details such as skills, name, email, and phone number. Using Precision, Recall, and F1 Score as metrics, the module achieved a Precision of 0.92, Recall of 0.93, and an F1 Score of 0.92, demonstrating its effectiveness in accurately extracting essential candidate information.

The evaluation of the Question Generation Module in FairHire was conducted by comparing Gemini-generated questions to 30 expert-written questions using ROUGE and METEOR metrics. The generated and reference questions were tokenized, and scores were computed to assess content overlap, fluency, and semantic accuracy. The system achieved a ROUGE-1 score of 0.79, ROUGE-2 of 0.65, ROUGE-L of 0.72, and a METEOR score of 0.81, demonstrating strong alignment with expert-designed content.

TABLE I
EVALUATION METRICS FOR QUESTION GENERATION

Metric	Score
ROUGE-1	0.79
ROUGE-2	0.65
ROUGE-L	0.72
METEOR	0.81

The evaluation of the Scoring Module in FairHire was conducted by comparing Gemini-assigned scores to those provided by two domain experts across 25 candidate responses. Responses were evaluated based on Relevance, Accuracy, and Depth using structured prompts. To measure performance, the system's scores were compared to expert scores using Mean Absolute Error (MAE) and Pearson's Correlation Coefficient. The module achieved an MAE of 0.42 and a correlation score of 0.87, indicating minimal deviation and strong alignment with expert evaluations.

The Proctoring Module in FairHire was evaluated using controlled scenarios that simulated suspicious behaviors such as tab switching, multiple faces, and mobile phone usage. The system achieved a TPR of 0.92, an FPR of 0.08, and

TABLE II
EVALUATION METRICS FOR RESPONSE SCORING

Metric	Value
Mean Absolute Error (MAE)	0.42
Pearson's Correlation	0.87

a Detection Latency (DL) of 450 ms, ensuring accurate and timely identification of malpractice during interviews.

The Speech Module in FairHire was evaluated using Word Error Rate (WER), BLEU Score, and Detection Latency to assess transcription accuracy and speed. The system achieved a WER of 0.08, a BLEU Score of 0.91, and an average latency of 280 ms, demonstrating highly accurate and efficient speech-to-text conversion, ensuring seamless performance during real-time interviews.

VI. CONCLUSION

FairHire is an advanced AI-driven system that aims to transform the recruitment process by providing an efficient, objective, and unbiased platform for candidate evaluation. The system integrates multiple modules, including Resume Profiling, Question Generation, Speech Interaction, Proctoring, and Scoring, each designed to ensure a personalized and thorough interview experience. Through Resume Profiling, it extracts key skills, qualifications, and experiences, which are then used to generate adaptive, role-specific interview questions that evaluate a candidate's depth of knowledge. The Speech Module facilitates real-time interaction, converting spoken responses into text for accurate evaluation, while the Proctoring Module ensures interview integrity by monitoring candidate behavior, detecting potential cheating or distractions. The Scoring Module assesses candidate responses based on predefined criteria and flags. Looking forward, FairHire could expand by incorporating more sophisticated AI techniques, such as emotion recognition and deeper integration with HR systems, to create a more holistic and streamlined hiring process that reduces bias and enhances decision-making.

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